

Normalisation

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Form 1: Port Tours Available

✓ **UNF**

PORT_TOUR_AVAILABLE (port_code, port_name, country_code, country_name, (temp_month, temp_avg_high, temp_avg_low), (tour_no, tour_name, tour_desc, tour_hours_req, tour_cost_pp, tour_wc_access, tour_availability, tour_start_time))

Repeating groups:

- temp_month, temp_avg_high, temp_avg_low
- tour_no, tour_name, tour_desc, tour_hours_req, tour_cost_pp, tour_wc_access, tour_availability, tour_start_time

1NF

✓ PORT_TOUR_AVAILABLE (port_code, port_name, country_code, country_name)

✓ TEMPERATURE (port_code, temp_month, temp_avg_high, temp_avg_low)

✓ TOUR_DETAILS (port_code, tour_no, tour_name, tour_desc, tour_hours_req, tour_cost_pp, tour_wc_access, tour_availability, tour_start_time)

✓ *Partial dependencies:*

- No partial dependencies

✓ **2NF**

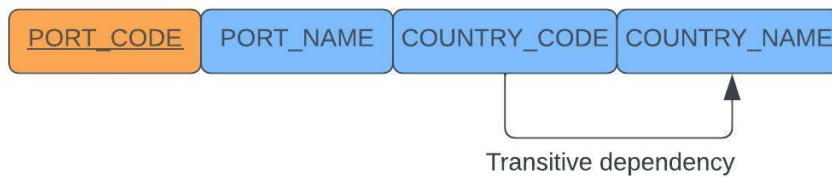
PORT_TOUR_AVAILABLE (port_code, port_name, country_code, country_name)

TEMPERATURE (port_code, temp_month, temp_avg_high, temp_avg_low)

TOUR_DETAILS (port_code, tour_no, tour_name, tour_desc, tour_hours_req, tour_cost_pp, tour_wc_access, tour_availability, tour_start_time)

✓ *Transitive dependencies:*

- country_code → country_name



✓ 3NF

COUNTRY (country_code, country_name)

PORT_TOUR_AVAILABLE (port_code, port_name, country_code)

TEMPERATURE (port_code, temp_month, temp_avg_high, temp_avg_low)

TOUR_DETAILS (port_code, tour_no, tour_name, tour_desc, tour_hours_req, tour_cost_pp, tour_wc_access, tour_availability, tour_start_time)

✓ Full dependencies

- country_code → country_name
- port_code → port_name, country_code
- port_code, temp_month → temp_avg_high, temp_avg_low
- port_code, tour_no → tour_name, tour_desc, tour_hours_req, tour_cost_pp, tour_wc_access, tour_availability, tour_start_time

Form 2: Port - Tour Participant Report

✓ UNF

PORT_TOUR_PARTICIPANT (port_code, tour_no, tour_name, tour_date, tour_start_time, (passenger_id, passenger_name, passenger_language, cruise_id, cruise_name, payment_received) *decompose*)

Repeating groups:

passenger_id, passenger_name, passenger_language, cruise_id, cruise_name, payment_received

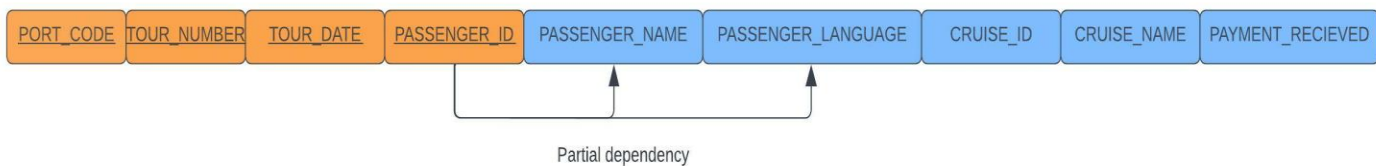
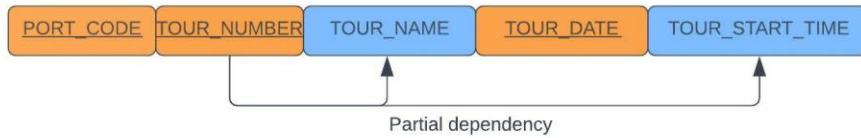
1NF

✓ PORT_TOUR_PARTICIPANT (port_code, tour_no, tour_name, tour_date, tour_start_time)

✓ REGISTRATION (port_code, tour_no, tour_date, passenger_id, passenger_name, passenger_language, cruise_id, cruise_name, payment_received)

✓ Partial dependencies:

- $\text{port_code, tour_no} \rightarrow \text{tour_name, tour_start_time}$
- $\text{passenger_id} \rightarrow \text{passengers_name, passenger_language}$



✓ 2NF

TOUR_DETAILS (port_code, tour_no, tour_name, tour_start_time)

PORT_TOUR_PARTICIPANT (port_code, tour_no, tour_date)

PASSENGER (passenger_id, passenger_name, passenger_language)

REGISTRATION (port_code, tour_no, tour_date, passenger_id, cruise_id, cruise_name, payment_received)

✓ Transitive dependencies:

- $\text{cruise_id} \rightarrow \text{cruise_name}$



✓ 3NF

TOUR_DETAILS (port_code, tour_no, tour_name, tour_start_time)

PORT_TOUR_PARTICIPANT (port_code, tour_no, tour_date)

PASSENGER (passenger_id, passenger_name, passenger_language)

REGISTRATION (port_code, tour_no, tour_date, passenger_id, cruise_id, payment_received)

CRUISE (cruise_id, cruise_name)

✓ Full dependencies

- port_code, tour_no → tour_name, tour_start_time
- port_code, tour_no, tour_date
- passenger_id → passenger_name, passenger_language
- port_code, tour_no, tour_date, passenger_id → cruise_id, payment_received
- cruise_id → cruise_name

Collected 3NF Relations from Both Forms

1. PORT_TOUR_AVAILABLE (port_code, port_name, country_code)
2. COUNTRY (country_code, country_name)
3. TEMPERATURE (port_code, temp_month, temp_avg_high, temp_avg_low)
4. TOUR_DETAILS (port_code, tour_no, tour_name, tour_desc, tour_hours_req, tour_cost_pp, tour_wc_access, tour_availability, tour_start_time)
5. TOUR_DETAILS (port_code, tour_no, tour_name, tour_start_time)
6. PORT_TOUR_PARTICIPANT (port_code, tour_no, tour_date)
7. PASSENGER (passenger_id, passenger_name, passenger_language)
8. REGISTRATION (port_code, tour_no, tour_date, passenger_id, cruise_id, payment_received)
9. CRUISE (cruise_id, cruise_name)

Attribute Synthesis

1.

✓ PORT (port_code, port_name, country_code) ✓

2.

✓ COUNTRY (country_code, country_name) ✓

3.

✓ TEMPERATURE (port_code, temp_month, temp_avg_high, temp_avg_low) ✓

4. & 5.

✓ TOUR (port_code, tour_no, tour_name, tour_desc, tour_hours_req, tour_cost_pp, tour_wc_access, tour_availability, tour_start_time) ✓

6.

✓ TOUR_DATE (port_code, tour_no, tour_date) ✓

7.

✓ PASSENGER (passenger_id, passenger_name, passenger_language) ✓ **Note:** Based on Assignment 1A, passenger_name will split into two attributes which are passenger_first_name and passenger_last_name.

8.

✓ REGISTRATION (port_code, tour_no, tour_date, passenger_id, cruise_id, payment_received) ✓

9.

✓ CRUISE (cruise_id, cruise_name) ✓